

KARTHIK RAMADUGU

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PROFESSIONAL SUMMARY

Versatile Software Developer with 2+ years of experience in full-stack development, cloud-native applications, and DevOps automation. Proficient in Node.js, React.js, Express.js, and MongoDB, with expertise in building scalable microservices and APIs. Skilled in AWS, Azure, GCP, CI/CD, Docker, Kubernetes, Terraform, and Ansible. Strong background in system performance optimization, monitoring, and Agile/Scrum collaboration. Currently pursuing an M.S. in Computer Science at the University of Dayton.

TECHNICAL SKILLS

Programming & Scripting: JavaScript (ES6+), TypeScript, Python, SQL, Shell/Bash .

Frontend Development: React.js (v18), Next.js, Redux, Tailwind CSS, Material UI, HTML5, CSS3, Chart.js, D3.js .

Backend Development: Node.js, Express.js, RESTful APIs, GraphQL, Microservices Architecture .

Databases: MongoDB, PostgreSQL, MySQL, Redis, Firebase, SQL Server .

Cloud Platforms: AWS (EC2, S3, Lambda, CloudFront, CloudWatch, RDS, EKS), Azure (App Services, Functions, Cosmos DB),GCP (Cloud Run, Firestore).

DevOps & Infrastructure: Docker, Kubernetes, Jenkins, GitHub Actions, GitLab CI/CD, Terraform, Ansible, Nginx, Prometheus, Grafana.

Testing & QA: Jest, Mocha, Chai, Cypress, Selenium, Postman.

Version Control & Collaboration: Git, GitHub, GitLab, Bitbucket, Jira, Confluence, Agile/Scrum.

UI/UX & Design Tools: Figma, Adobe XD, Responsive Design, Wireframing, Prototyping, Usability Testing.

APIs & Integration: Swagger/OpenAPI, JSON, XML, WebSockets, Event-Driven Architecture.

PROFESSIONAL EXPERIENCE

Software Engineer Intern

Jun 2024 -Jul 2025

CAS | Columbus, OH

- Engineered REST APIs and microservices in Node.js, Express.js, and MongoDB, which cut query response times by 30% for scientific data retrieval.
- Automated AWS deployments with EC2, Lambda, and S3 while setting up CloudWatch alerts, driving 35% faster release cycles and stable rollouts.
- Designed CI/CD pipelines through Jenkins and GitHub Actions that supported continuous delivery, ensuring production updates shipped with zero downtime.
- Improved backend throughput by refining MongoDB queries and layering Redis caching, reducing data delays by 25% under heavy workloads.
- Partnered with researchers and developers using Jira and Confluence, streamlining sprint planning and boosting velocity by 20%.
- Strengthened observability through Prometheus metrics and Grafana dashboards, which shortened anomaly detection and cut recovery time by 35%.
- Built unit and integration suites in Jest and Supertest, lifting test coverage to 85% and reducing post-release issues.
- Refined service code with ESLint standards and modular patterns, improving maintainability and helping onboard new engineers faster.

Trainee Software Engineer

Nov 2022 - May 2023

FunctionUp | Bengaluru, India

- Delivered backend APIs on Node.js and Express.js that handled client requests more efficiently, reducing response latency by 30%.
- Introduced Mocha and Chai testing frameworks, halving manual QA time while lowering deployment rollbacks by 25%.
- Configured AWS EC2, RDS, and S3 with load balancing, raising system uptime from 90% to 99.5% during stress scenarios.
- Produced detailed Swagger/OpenAPI documentation, accelerating developer onboarding and cutting integration issues by 25%.
- Contributed to Agile ceremonies including sprint reviews and retrospectives, helping the team consistently achieve a 95% sprint completion rate.
- Automated deployments using GitHub Actions CI/CD workflows, trimming manual work by 40% and reducing release delays.
- Enabled proactive error handling by setting up CloudWatch alarms and structured logs, which cut downtime by 20%.
- Rewrote legacy modules with ESLint compliance and modular architecture, enhancing code readability and lowering new engineer ramp-up time by 30%.

PROJECTS

IndiaMart - E-Commerce Platform

- Built a full MERN stack platform (React, Node.js/Express, MongoDB) deployed on AWS EC2 with S3 and load balancing, integrating Redis caching to cut API latency by 45% while supporting 10K+ concurrent sessions.
- Developed real-time inventory and order synchronization using WebSockets and Redis pub/sub, which reduced stockouts by 60% during peak shopping traffic.
- Implemented server-side collaborative filtering with Node.js and Redis feature store, delivering personalized recommendations in a React + TailwindCSS UI, boosting retention and conversions by ~35%.

Fitness Tracker - Nutrition & Workout Companion

- Created a MERN application with JWT and OAuth2 authentication, deploying frontend on Vercel and backend on Render autoscaling, ensuring 95%+ uptime with secure token and data storage.
- Designed personalized dashboards using React, Chart.js, and serverless Node.js microservices, which raised goal completion rates by 25% among active users.
- Introduced gamification features with Firebase push notifications and activity streaks, driving a 40% increase in daily active engagement and longer session times.

URL Shortener - Smart Link Management

- Architected a stateless Node.js API with sharded MongoDB and Dockerized workers, deployed on AWS EC2 Auto Scaling, sustaining 50K+ links/month with 99.9% uptime.
- Built React + Chart.js analytics dashboards aggregating clickstream and referral metrics, enabling data-driven insights that improved click-through rates by 25%.
- Strengthened platform security with JWT authentication, rate limiting, and custom domain support, reducing abuse by 35% while enabling enterprise- grade branding.

EDUCATION

Master of Science in Computer Science

Expected Dec 2025

University of Dayton | Dayton, OH

CERTIFICATIONS

- Cloud Practitioner Essentials - **Coursera**
- AWS Docker and Kubernetes for Developers - **Coursera**
- DevOps Foundations: Continuous Delivery/Continuous Integration - **LinkedIn Learning**
- React.js Essential Training - **LinkedIn Learning**